The Media Connectivity Association



OVERVIEW AND UPDATES · NON-TERRESTRIAL NETWORKS (NTN)

Non-Terrestrial Networks in 5G Systems

Expanding network connectivity via satellite and HAPS for content delivery

3GPP

OCTOBER 2026 – www.5g-mag.com

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About the project

The documents the project rests on

3GPP TR 38.811

Technical Report

The foundational NTN study item: the study on NR access to non-terrestrial networks, completed in Release 15.

3GPP TR 38.821

Technical Report

Solutions for NR to support non-terrestrial networks, the outcome of the Release 16 study on solutions for NR for NTN.

3GPP TS 38.300

Technical Specification

The NR overall description. Its NTN aspects define the feeder and service links, the payload types and the cell types.

FIND OUT MORE **Technical Analysis** **Specifications**

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What NTN changes

NTN reuses the 5G system and the NR protocol stack; the differences are in the radio and in a few architecture roles

System model

where the NTN payload sits

- Service link
- Feeder link
- NTN Gateway
- Transparent payload
- Regenerative payload
- GNSS-capable UE

Cells

three types of service link

- Earth-fixed
- Quasi-Earth-fixed
- Earth-moving

Radio

long delay and large Doppler shift

- Timing advance
- Common timing advance
- Doppler pre-compensation
- k-offset
- Adapted HARQ, RRC and MAC timers

System information

assistance for the UE

- SIB19: satellite assistance
- SIB27: intended service areas of MBS broadcast

The NTN payload is on board a satellite or a high altitude platform station. SIB19 comes from the Release 17 NTN work and SIB27 from Release 19.

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How it grew

The 3GPP work on non-terrestrial networks, release by release

Rel. 15	Rel. 16	Rel. 17	Rel. 18	Rel. 19	Rel. 20
A study on NR for non-terrestrial networks (TR 38.811).	Studies TR 38.821 and TR 22.822. Only stage 1 of 5GSAT was defined.	First normative NTN: satellite RAT types, NR over transparent payloads with SIB19, NB-IoT and eMTC.	Satellite backhaul, discontinuous coverage, FR2-NTN bands and NTN-TN mobility.	Broadcast with an intended service area under one cell (SIB27), gNB on board, Store and Forward.	Release 20 (work plan of 28 September 2026), unverified: 5GSAT_Ph4, IoT_NTN_Ph4; TR 38.863 adds 3 MHz for n250 to n253.

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Mobility and MBS over NTN

What Release 19 specifies, as analysed on the Technical Analysis pages

Mobility in a single NTN operator

- Time- and location-based measurement initiation
- Conditional handover: A4, T1, D1, D2
- Satellite switch with re-synchronization
- Feeder link switchover

Deployment models the analysis's own grouping

- Single NTN operator
- Common NTN and TN operator
- Independent NTN and TN operators

MBS broadcast the Intended Service Area

- 5G Core to NG-RAN
- SIB27
- MCCH
- UE behaviour

MBS multicast NG-RAN decides

- PTP and PTM switching
- Multicast DRX HARQ timer for NTN
- Mobility of receivers

MBS multicast has no NTN-specific part in TS 23.247. For independent operators, MBS roaming is not supported in Release 19; Release 20 supports services of the visited network only.

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